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Progetto Melampo

From Attention to Intuition

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Scientific consolidation treatise for a neurobiological, multimodal, and theoretical-quantum architecture of assisted clinical intuition

Abstract

Project Melampo is consolidated here as a unified scientific program for clinical *Computer-Aided Intuition*. The core premise is that next-generation assisted diagnosis cannot be reduced either to pattern classification or to text generation alone. What is required is an architecture in which **3D perception**, **episodic and semantic memory**, **differential reasoning**, **metacognition**, **uncertainty quantification**, **clinical interoperability**, and **regulatory validation** operate as an integrated system [1–9, 13–20, 25–38].

The treatise adopts four structural principles.

- **Neurobiological principle.** Thought emerges from distributed networks: working memory, associative networks, the default mode network, predictive processing, replay, and neuromodulation must be translated into observable computational modules [1–6].
- **Pedagogical-metacognitive principle.** Clinical expertise does not coincide with a single inferential act; it is built through practice, reflection, error control, hypothesis revision, and progressive learning [7–12].

- **Engineering principle.** The most promising state of the art, updated to April 2026, points toward federated stacks: 3D volumetric encoders, medical vision-language foundation models, knowledge graphs and retrieval, model orchestrators, generative replay, risk calibrators, interaction protocols, and Total Product Life Cycle governance [13–38].
- **Theoretical-quantum principle.** The theoretical-quantum hypothesis is not removed from the project. It is integrated as a formal and falsifiable research line articulated across three coupled levels: **quantum-like cognition**, **open Schrödinger/Lindblad-type dynamics**, and a **biophysical frontier** involving microtubules, radical pairs, and other non-classical mechanisms in biological systems [39–48].

The central thesis is that Melampo should operate as a **multi-layer clinical architecture**: an immediately implementable perceptual-grounded layer; a cognitive-metacognitive layer governing differential diagnosis, escalation, and abstention; and a theoretical-quantum layer modeling contextuality, evidence order, hypothesis interference, and, at the frontier, possible non-classical biophysical substrates. None of these levels is self-sufficient. The robustness of the project depends on their **experimentally testable interaction**.

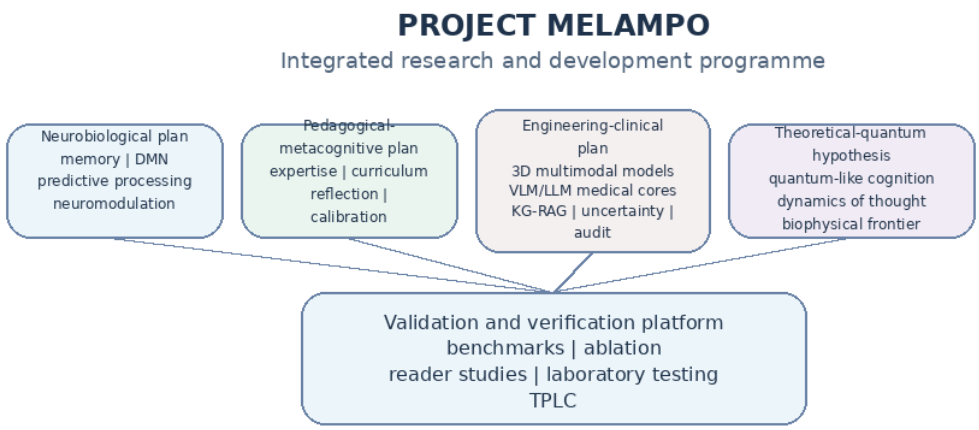

Figure 1. Integrated planes of Project Melampo.

1. Methodological Premise and Scientific Positioning

Melampo begins from a strong intuition: the value of an advanced diagnostic system does not coincide with average accuracy alone, but with the ability to compose **heterogeneous evidence**, manage **ambiguity**, know **when not to issue a judgment**, and restructure its hypotheses when new information arrives. A system of this kind is neither a simple classifier nor a simple medical chatbot. It is an architecture for assisted clinical reasoning.

This document therefore adopts a precise stance. Melampo is not presented as a project fragmented into unrelated parts, but as a **theoretical-operational continuum** in which different levels are distinguished only for methodological clarity.

Two scientific obligations follow. The first is to translate every major claim into **verifiable constructs**. The second is **not** to amputate frontier hypotheses prematurely, but to submit them to a program of differential testing, comparison with rival models, and explicit criteria for promotion, suspension, or rejection.

2. Neurobiology of Thought and Cognitive Processing

2.1 Distributed networks, working memory, and executive control

Contemporary evidence shows that working memory is not a static container but a distributed circuit centered on prefrontal cortex and supported by recurrent excitatory-inhibitory dynamics, inter-areal synchronization, and dopaminergic modulation [2]. In design terms, this implies that Melampo should not contain a single cognitive “center,” but rather a **diagnostic workspace** in which images, reports, history, ontological knowledge, and the current differential state remain simultaneously accessible across several inferential cycles.

A minimal classical formalization of the neuroelectrical substrate, useful as an observable macroscopic layer, is given by the Hodgkin–Huxley equation:

$$C_m \frac{dV}{dt} = I_{\text{ext}} - \bar{g}_{\text{Na}} m^3 h(V - E_{\text{Na}}) - \bar{g}_{\text{K}} n^4 (V - E_{\text{K}}) - g_L (V - E_L).$$

Where:

- C_m is the membrane capacitance;
- V is the membrane potential;
- I_{ext} is the external current;
- $\bar{g}_{\text{Na}}, \bar{g}_{\text{K}}, g_L$ are maximal or leakage conductances;
- m, h, n are gating variables;
- $E_{\text{Na}}, E_{\text{K}}, E_L$ are ionic equilibrium potentials.

In this treatise the equation serves an important function: it establishes that Melampo's observable neural level remains compatible with a **classical electrophysiological description**. The theoretical-quantum hypothesis does not replace that level; it subsumes it as a plane of macrobiological observables upon which effective parameters and experimental comparisons can be defined.

2.2 Default mode network, spontaneous thought, simulation, and abstraction

The default mode network has been redefined as a network active in memory, abstraction, scenario simulation, and the integration of internal and external information [1]. More recent evidence further suggests that spontaneous thought emerges from dynamics distributed beyond the DMN alone, involving network configurations that change with the content of mental flow [4].

For Melampo this is decisive: the intuitive component should not be treated as an obscure event, but as a **rapid compression of prior experience, abstract schemas, and contextual signals**. The system should therefore include an offline or quasi-offline mode devoted to trace reactivation, counterfactual construction, and the combination of distant patterns.

2.3 Predictive processing, global workspace, and surprise

Predictive processing interprets the brain as a generative hierarchy that formulates predictions and corrects its internal models via prediction errors [3, 49]. The corresponding computational translation for Melampo is straightforward: every new examination, symptom, or radiological finding changes the ranking of the differential diagnosis because it changes free energy, surprise, and the precision assigned to information channels.

A compact formulation of the inferential objective is:

$$\mathscr{F}_t = \mathbb{E}_{q(s_t)}[\log q(s_t) - \log p(o_t, s_t \mid M)].$$

Where:

- $\mathscr{F}_t$ is the free energy at time t;
- $q(s_t)$ is the approximate distribution over latent clinical states;
- o_t are the observations available at step t;
- M is the structured memory of the system;
- $p(o_t, s_t \mid M)$ is the joint generative model of observations and states.

In Melampo, minimizing $\mathscr{F}_t$ is not a philosophical abstraction. It defines the criterion by which the diagnostic workspace integrates new evidence and decides whether to investigate further, abstain, or consolidate a hypothesis.

2.4 Replay, sleep, dreaming, and constructive consolidation

Hippocampal replay during sleep and the microstructure of sleep are now strongly associated with memory consolidation [5]. A generative model of memory construction and consolidation has also shown that replay can be interpreted as the training of generative systems capable of reconstructing and recombining experience [6]. This passage is crucial for the project's Section 7.2: Melampo's **statistical dream** is not ornamental metaphor, but a computational strategy inspired by replay and constructive consolidation.

Accordingly, the generative module must:

- reactivate real cases of high salience;
- synthesize plausible cases from latent regions conditioned by clinical context;
- produce hard negatives and borderline cases;
- discard incoherent trajectories;
- consolidate only those samples that improve discrimination, calibration, or rare-case coverage.

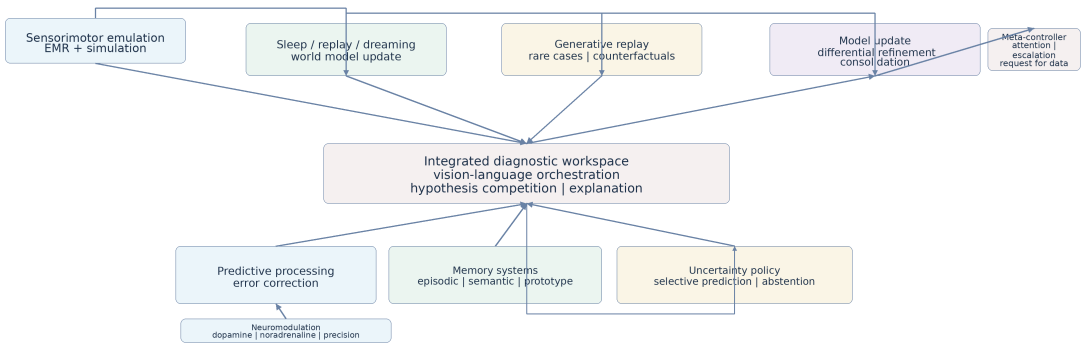

Figure 2. Neurocognitive map and corresponding Melampo modules.

2.5 Neuromodulation as computational control

The role of dopamine and noradrenaline should not be reduced to mere metaphor. Recent literature continues to show that dopaminergic and adrenergic systems modulate learning, motivation, signal precision, temporal regulation of synaptic output, and decision under uncertainty [50–52]. In Melampo they are translated into **computational control variables**: precision, volatility, update rate, novelty threshold, and abstention threshold.

A minimal biophysical relation between calcium and synaptic release probability can be written as:

$$p_{\text{rel}} = 1 - \exp\left(-k[\text{Ca}^{2+}]^{n}\right).$$

Where:

- p_{rel} is the synaptic release probability;
- k is a scale constant;
- $[Ca^{2+}]$ is intracellular calcium concentration;
- n is the cooperativity exponent.

Within the project, observable neurochemical quantities function as a bridge toward a more general statistical dynamics: they do not coincide with the wavefunction, but can modulate effective parameters governing stability, decoherence, precision, and decisional commitment.

3. Evolution of Thought in Pedagogy, Learning, and Clinical Expertise

The construction of intuition is not only neurobiological. It is also pedagogical. Piaget describes the emergence of cognitive structures through progressive organization; Vygotsky shows the decisive role of social mediation and cultural tools; Bruner emphasizes the function of representations and the narrative construction of meaning [10–12]. In clinical domains, expertise emerges from the intertwining of:

- real-case experience;
- dialogue with teachers and peers;
- retrospective correction of error;
- internalization of diagnostic schemas and counterexamples;
- metacognitive reflection on one's own reliability.

In the neuro-educational literature and recent medical pedagogy, metacognition is defined as the capacity to monitor and regulate one's own cognitive processes [7–8]. For Melampo this implies an architecture with two obligations:

- an obligation of **self-monitoring**, that is, explicit estimation of safety, coverage, and limits;
- an obligation of **self-regulation**, that is, the ability to request additional data, activate retrieval, expand the differential diagnosis, or abstain.

The debate on dual-process models of clinical reasoning also suggests caution: diagnostic quality depends primarily on the nature and availability of retrieved knowledge, not on a rigid opposition between an intuitive system and an analytical one [9]. The design solution is therefore a **dynamic continuum** between rapid response and slower revision, mediated by uncertainty thresholds and risk cost.

4. State of the Art of Multimodal Medical AI as of April 11, 2026

4.1 General trend line

The most promising trajectory in medical AI does not converge toward a single monolithic model, but toward **coordinated ecosystems** of specialist and generalist models. Medical foundation models have become more relevant because they combine large-scale pretraining, adaptability across multiple clinical tasks, and integration across text, images, longitudinal data, and knowledge sources [13]. In parallel, 2025–2026 accelerated three frontier lines:

- vision-language models for 3D imaging [15, 19];
- multimodal foundation models for pathology on whole-slide images [17–18];
- orchestration of multiple models through standardized interaction protocols [20–24].

4.2 Volumetric encoders and 3D self-supervised learning

For Melampo's perceptual core, the most robust families remain transformer-based volumetric encoders or hybrid transformer-CNN stacks. 3D Swin Transformer, Swin-UNETR, and their descendants opened the path, while more recent work on 3D self-supervised learning indicates that

generalization increasingly depends on pretraining quality and multimodal anatomical coverage [15–16]. Architecturally, this leads to a preference for:

- 3D encoders pretrained on large CT/MRI collections;
- multi-resolution volumetric tokenization;
- support for multi-view and whole-volume context;
- lightweight adaptation (adapters, LoRA, prompt tuning) for site-specific deployment.

4.3 Medical vision-language foundation models and the most promising models

By 2025, the review literature on vision-language foundation models for 3D imaging had documented rapid growth in systems combining volumetric backbones with LLMs or language decoders for report generation, VQA, and multimodal reasoning [15]. In 2026 this axis was further enriched by models such as **Decipher-MR**, explicitly designed for 3D MRI representations, and the **MedGemma 1.5** update presented by Google as an open medical model capable of handling text, 2D images, high-dimensional volumes, and whole-slide pathology [19, 53].

For Melampo this implies a precise architectural choice: do not entrust everything to a single VLM, but construct a **federation of modules**.

4.4 Multifactoring, real multimodality, and interaction among models

The term *multifactoring* is used here in a technical sense: a serious clinical decision requires the fusion of heterogeneous factors—morphology, symptoms, laboratory data, chronology, provenance, therapeutic history, rare patterns, literature, guidelines, and local resources. The optimal model is not the one that merely sees more, but the one that can **align heterogeneous factors within a verifiable update structure**.

A useful formalism for cross-modal fusion is:

$$Z_t = \text{Attn}(Q_t, K_t, V_t) = \text{softmax}\left(\frac{Q_t K_t^\top}{\sqrt{d_k}}\right) V_t.$$

Where:

- Q_t is the active clinical query at step t;
- K_t and V_t are keys and values produced by the various encoders (visual, textual, knowledge-based);
- d_k is the normalization dimensionality;
- Z_t is the fused multimodal representation.

When modules are multiple, Melampo should use an orchestrator. A simple mixture-of-experts formulation is:

$$\hat{y} = \sum_{j=1}^{J} \pi_j(x) f_j(x), \qquad \pi_j(x) = \frac{\exp(g_j(x)/\tau)}{\sum_{\ell=1}^{J} \exp(g_\ell(x)/\tau)}.$$

Where:

- $f_j(x)$ is the contribution of expert j;
- $g_j(x)$ is the router score for expert j;
- $\pi_j(x)$ is the weight assigned to that expert;
- τ is the gating temperature;
- $\hat{y}$ is the integrated output.

This formulation is especially suitable for Melampo when combining a 3D perceptual specialist, a semantic specialist for EHR and guidelines, a retriever/knowledge-graph component, and a generator/verifier.

MELAMPO ARCHITECTURAL STACK

From grounded multimodal perception to metacognitive control

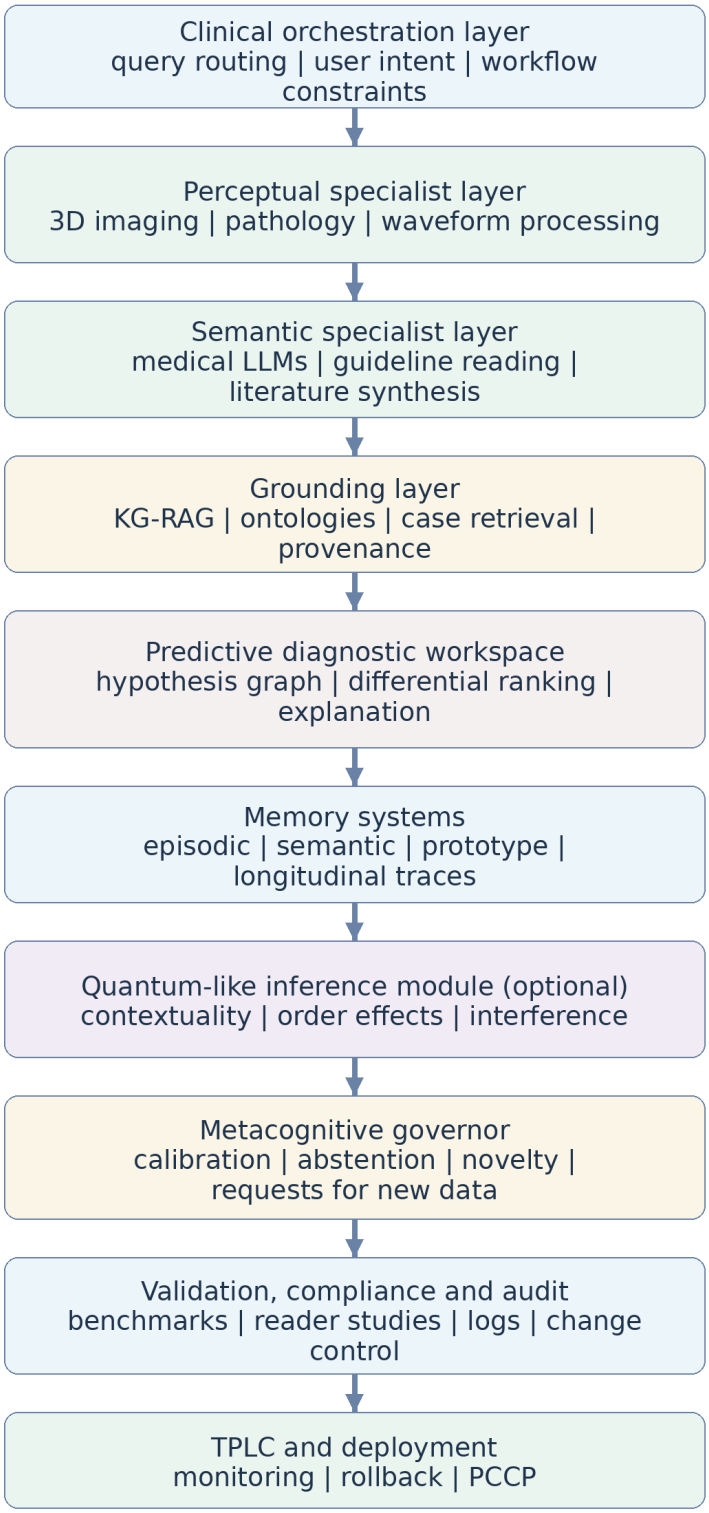

Figure 3. Melampo architectural stack.

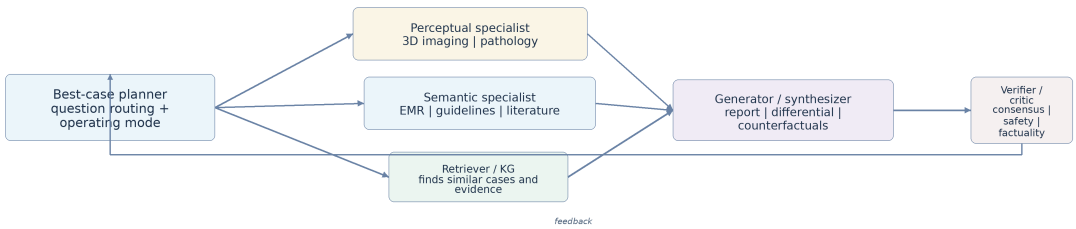

Figure 4. Orchestration of specialist models.

4.5 Languages for model interaction and semantic standards

If Melampo is to become a platform rather than an isolated prototype, it needs two families of interaction languages.

The first concerns the **clinical domain**:

- FHIR for clinical resources and interoperable workflow [23];
- DICOM / DICOM-SR for images, metadata, and structured reports [24];
- SNOMED CT, LOINC, and ICD for shared semantics.

The second concerns **cooperation among models and tools**:

- the **Model Context Protocol (MCP)** as an open standard for connecting models, tools, resources, and context via JSON-RPC [21];
- **Agent2Agent (A2A)** as a protocol for collaboration among agents/models from different vendors and frameworks [22];
- JSON schemas and clinical ontologies for structured, validable, and signable outputs.

The emerging picture is clear: Melampo's future is not an isolated LLM, but a **multi-model clinical system** reasoning over established healthcare standards and open orchestration protocols.

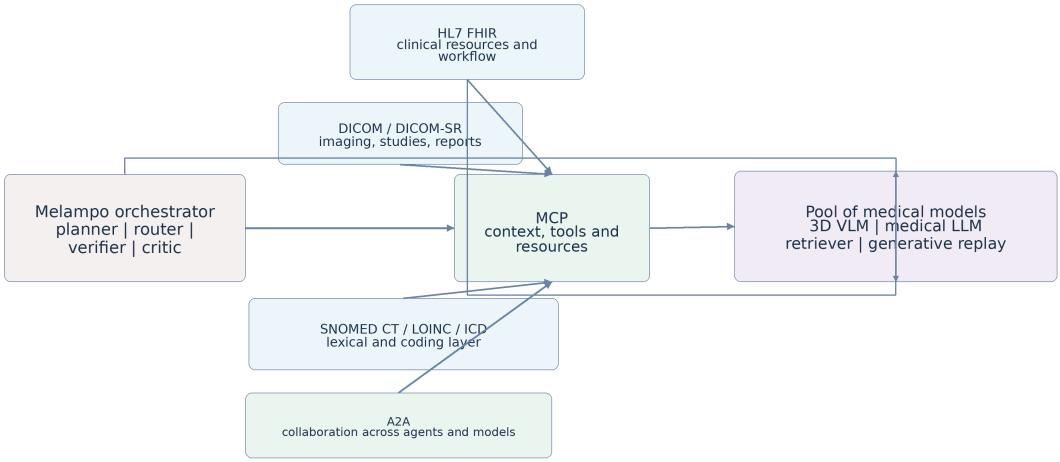

Figure 5. Ecosystem of protocols and standards for Melampo.

4.6 KG-RAG, structured memory, and grounding

Recent reviews of RAG in healthcare converge on one point: grounding, transparency, and updateability improve when the model can retrieve external evidence, but the benefits critically depend on document quality, retrieval quality, and post-retrieval reasoning [25]. In Melampo, knowledge graphs and RAG must therefore be structural parts of semantic memory, not optional accessories.

A composite retrieval score can be expressed as:

$$S(d,q) = \alpha S_{\text{dense}}(d,q) + \beta S_{\text{sparse}}(d,q) + \gamma S_{\text{KG}}(d,q) + \delta S_{\text{site}}(d,q).$$

Where:

- d is the retrieved document or case;
- q is the clinical query;
- S_{dense} is the dense semantic score;
- S_{sparse} is the lexical/sparse score;
- S_{KG} measures coherence with the knowledge graph;
- S_{site} weights local or institutional relevance;
- $\alpha, \beta, \gamma, \delta$ are calibrable weights.

4.7 Generative replay, case synthesis, and continual training

Synthetic generation of clinical and imaging data has made a qualitative leap. Reviews from 2025–2026 describe a landscape dominated by diffusion models, VAEs, GANs, LLMs, and multimodal systems, with applications ranging from image synthesis to tabular/EHR synthesis, longitudinal data generation, and data balancing [31–33]. At the same time, the literature on continual learning in medicine has clarified that, without replay or other constraints, the risk of catastrophic forgetting is high [30].

Melampo should therefore use the generative module in three modes:

- controlled augmentation of rare classes;
- simulation of counterfactual cases and temporal transitions;
- statistical dreaming for reorganizing memory and differential diagnosis.

A conditional VAE loss suitable for replay is:

$$\mathscr{L}_{\text{VAE}} = \mathbb{E}_{q_\phi(z \mid x, c)}[-\log p_\theta(x \mid z, c)] + \beta D_{\text{KL}}(q_\phi(z \mid x, c) \| p(z \mid c)).$$

Where:

- x is the observed case (image, text, or multimodal input);
- c is the conditioning clinical context;
- z is the latent variable;
- q_ϕ is the variational encoder;
- p_θ is the decoder/generator;

- $p(z \mid c)$ is the context-conditioned prior;
- β controls the weight of KL regularization.

4.8 Uncertainty, abstention, and clinically serious systems

Recent literature on uncertainty quantification in medical imaging distinguishes at least three families of uncertainty: **aleatoric**, **epistemic**, and **operational**. For Melampo, a fourth family must be added: **retrieval-grounding uncertainty** [26–27].

A useful operational decomposition is:

$$U_{\text{tot}} = U_{\text{alea}} + U_{\text{epi}} + U_{\text{retr}} + U_{\text{shift}}.$$

Where:

- U_{alea} is intrinsic data uncertainty;
- U_{epi} is model/parametric uncertainty;
- U_{retr} is uncertainty due to incomplete retrieval or grounding;
- U_{shift} measures dataset shift or out-of-distribution conditions.

A minimal emission policy is:

$$\text{emit}(t) = \mathbb{1}[\max_i p(d_i \mid o_t) \geq \tau_{\text{conf}} \wedge U_{\text{tot}} \leq \kappa].$$

Where:

- $p(d_i \mid o_t)$ is the probability of diagnosis d_i given observations o_t;
- τ_{conf} is the minimum confidence threshold;
- κ is the maximum tolerable risk threshold;
- $\mathbb{1}[\cdot]$ is the binary emission indicator.

5. Unified Scientific Architecture of Melampo

The proposed architecture is composed of seven tightly coupled functional layers.

5.1 Data layer and clinical normalization

Inputs enter Melampo through an ingestion layer handling 2D/3D DICOM, whole-slide pathology, EHR, reports, laboratory data, and therapeutic timelines. Semantic normalization produces FHIR resources, DICOM-SR objects, and terminology mappings onto SNOMED/LOINC/ICD.

5.2 Perceptual layer and cross-modal fusion

This layer hosts 3D encoders, pathology encoders, text encoders, and context adapters. Fusion must be multi-resolution and cross-modal, with both local and global evidence handling.

5.3 Memory and differential diagnosis

Melampo's memory is tripartite:

- **episodic**, containing analogous cases and known outcomes;
- **semantic**, containing guidelines, literature, ontologies, and protocols;
- **prototypical**, containing diagnostic clusters and rare variants.

A basic prototypical classifier may be written as:

$$c_k = \frac{1}{|S_k|} \sum_{x_i \in S_k} \phi(x_i), \qquad p(y = k \mid x) \propto \exp\left(-d(\phi(x), c_k)\right).$$

Where:

- S_k is the set of examples for class or prototype k;
- $\phi(x)$ is the embedding of case x;
- c_k is the prototypical center of class k;
- $d(\cdot,\cdot)$ is a latent-space distance.

5.4 Meta-controller, critique loop, and safety governor

The metacognitive layer evaluates whether the available data are sufficient, whether the differential should be expanded, whether retrieval should be activated, or whether clarification requests should be generated. The critique loop acts as an internal judge of clinical coherence, image-text correspondence, guideline adherence, and pathophysiological plausibility.

5.5 Generative replay and reorganization layer

The generator is not meant to “novelize” the case, but to produce clinically constrained replay and counterfactuals. This layer feeds both memory and differential reasoning. Its essential constraint is that every synthetic case be accepted only if it passes tests of clinical coherence and training utility.

5.6 Optional theoretical-quantum layer

The theoretical-quantum layer comes into play when evidence order, contextual ambiguity, or hypothesis interference render additive/classical models insufficient. It does not replace classical models; it accompanies them and challenges them experimentally.

6. Technical-Mathematical Formalization of the Implementable Core

6.1 Differential update

In the classical regime, Bayesian updating of the differential remains the most transparent starting point:

$$p(d_i \mid o_{1:t}) = \frac{p(o_t \mid d_i, o_{1:t-1})\, p(d_i \mid o_{1:t-1})}{\sum_j p(o_t \mid d_j, o_{1:t-1})\, p(d_j \mid o_{1:t-1})}.$$

Where:

- d_i is diagnostic hypothesis i;
- $o_{1:t}$ is the observation sequence up to time t;
- $p(d_i \mid o_{1:t-1})$ is the prior updated at the previous step;
- $p(o_t \mid d_i, o_{1:t-1})$ is the contextual likelihood.

In Melampo, this equation is the classical reference against which the benefits of quantum-like formalisms must be measured.

6.2 Continual learning and parameter consolidation

To avoid catastrophic forgetting during incremental updates, an Elastic Weight Consolidation term may be written as:

$$\mathscr{L}_{\text{EWC}} = \mathscr{L}_{\text{task}} + \frac{\lambda}{2} \sum_i F_i (\theta_i - \theta_i^\star)^2 .$$

Where:

- $\mathscr{L}_{\text{task}}$ is the loss of the new task;
- λ is the regularization coefficient;
- F_i is the Fisher importance of parameter θ_i;
- $\theta_i^\star$ is the value consolidated from the previous task.

6.3 Prediction quality and calibration

A standard global calibration metric is the Expected Calibration Error:

$$\text{ECE} = \sum_{b=1}^{B} \frac{|B_b|}{n} |\text{acc}(B_b) - \text{conf}(B_b)|.$$

Where:

- B_b is confidence bin b;
- $|B_b|$ is the number of examples in that bin;
- n is the total number of examples;
- $\text{acc}(B_b)$ is empirical accuracy in the bin;
- $\text{conf}(B_b)$ is mean predicted confidence in the bin.

Melampo's objective is not only to reduce error, but to reduce the gap between real accuracy and declared confidence.

7. Integrated Theoretical-Quantum Hypothesis

7.1 Logical foundation: contextuality, evidence order, and interference

Recent literature on quantum cognition has reinforced two points. First, quantum formalisms are useful already at the cognitive level for describing **order effects**, **response replicability**, **contextuality**, and **interference between alternatives** [39–41]. Second, realistic cognitive measurement tools are not always reducible to projective measurements; in some cases they require more general state-update maps [40].

For Melampo this means that the diagnostic differential can be treated as a **competitive belief state** in which the order of incoming evidence changes the shape of the update. This does not abolish Bayes; it defines a measurable theoretical competitor.

7.2 Before measurement, during measurement, after measurement

The project's crucial requirement can be formalized in three temporal stages.

7.2.1 Before measurement: coherent evolution of hypotheses

Before decisive new evidence arrives, the latent state of the hypotheses is represented by an effective wavefunction $|\Psi(t)\rangle$ evolving according to the Schrödinger equation:

$$i\hbar_{\text{eff}} \frac{\partial}{\partial t} |\Psi(t)\rangle = \hat{H}(t) |\Psi(t)\rangle.$$

Where:

- $|\Psi(t)\rangle$ is the superposition state of the diagnostic hypotheses at time t;
- $\hbar_{\text{eff}}$ is an effective scale constant of the formalism;
- $\hat{H}(t)$ is the effective Hamiltonian incorporating context, memory, and dynamic priors.

This equation expresses the following principle: before decisive clinical measurement, hypotheses compete continuously and deterministically in their state space. In other words, Melampo maintains a **controlled superposition of the diagnostic differential**.

7.2.2 During measurement: contextual selection

When a new high-information piece of evidence arrives—an exam, an imaging finding, a selective answer to a question, a laboratory marker—the state is subjected to measurement. The probability of diagnostic outcome d_i is given by Born's rule:

$$p(d_i \mid c_t) = \langle \Psi_t | \hat{\Pi}_i(c_t) | \Psi_t \rangle = \mathrm{Tr}(\hat{\Pi}_i(c_t)\, \rho_t).$$

Where:

- $\hat{\Pi}_i(c_t)$ is the measurement operator associated with diagnosis d_i in context c_t;
- ρ_t is the density operator equivalent of state $|\Psi_t\rangle$;
- $\mathrm{Tr}(\cdot)$ is the operator trace.

In Melampo, collapse is therefore interpreted as the **deterministic solution of a high-dimensional probabilistic problem**: overlapping symptoms, medical statistics, partial reports, and conflicting patterns converge toward a selected outcome. In this operational sense, collapse may be called **clinical intuition**, provided that it remains traceable, parameterized, and verifiable.

7.2.3 After measurement: consolidation or reopening of the differential

After measurement, the system must choose between two routes:

- consolidate the dominant hypothesis in memory if evidence exceeds coherence and risk thresholds;
- reopen the superposition by asking new questions or launching new retrieval operations if risk remains high.

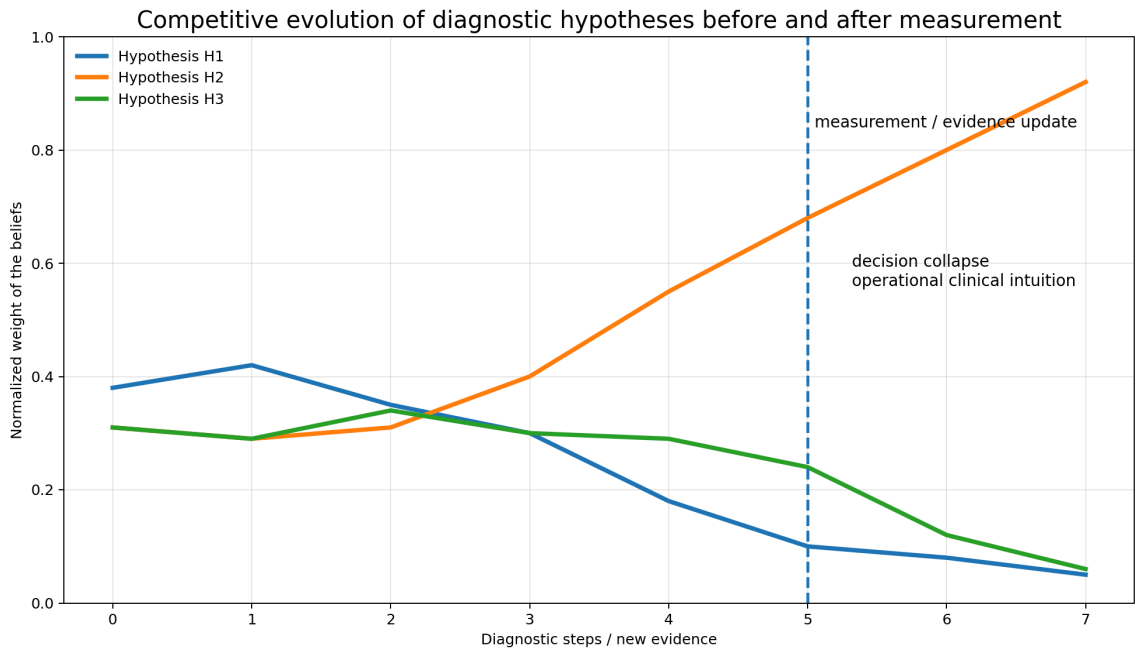

Figure 6. Competitive dynamics of hypotheses before and after measurement.

7.3 Open quantum systems: the Lindblad equation

Because the cognitive system is not isolated but continuously interacts with data, memory, noise, and clinical context, the most appropriate formulation is that of an open system:

$$\dot{\rho}_t = -\frac{i}{\hbar_{\text{eff}}}[\hat{H}(t), \rho_t] + \sum_j (\hat{L}_j \rho_t \hat{L}_j^\dagger - \frac{1}{2}\{\hat{L}_j^\dagger \hat{L}_j, \rho_t\}).$$

Where:

- ρ_t is the density operator of the diagnostic state;
- $[A,B] = AB - BA$ is the commutator;
- $\hat{L}_j$ are Lindblad operators modeling measurement, noise, retrieval, clinical constraints, or commitment;
- $\{A,B\} = AB + BA$ is the anticommutator.

This equation makes it possible to incorporate decoherence, contextual updating, interaction with the clinical environment, and safety thresholds without abandoning the project's formal structure.

7.4 Dirac-like formulation for coupled cognitive channels

The appeal to Dirac's equation is intended here as an effective formalism for describing two coupled cognitive channels—for example a fast channel and a deliberative channel, or a perceptual channel and a semantic channel:

$$i\hbar_{\text{eff}}\,\partial_t\,\Psi = [c_{\text{eff}}\,\boldsymbol{\alpha}\cdot\mathbf{p} + \beta m_{\text{eff}} c_{\text{eff}}^2 + V_{\text{ctx}}(t)]\Psi, \qquad \Psi = \begin{bmatrix} \psi_{\text{fast}} \\ \psi_{\text{slow}} \end{bmatrix}.$$

Where:

- Ψ is the two-component cognitive spinor;
- ψ_{fast} and ψ_{slow} represent two coupled channels;
- $\mathbf{p}$ is the momentum operator in latent space;
- $\boldsymbol{\alpha}$ and β are matrices of the Dirac formalism;
- c_{eff} is an effective velocity of information propagation;
- m_{eff} is a parameter of cognitive inertia;
- $V_{\text{ctx}}(t)$ is the context potential.

In this interpretation, the Dirac-like form imposes first-order dynamics and a coupling structure that may be useful for modeling fast-to-analytic transitions without falling back into rigid dualism.

7.5 From neurochemistry to non-classical statistical dynamics

The bridge required by the project between chemistry, electrophysiology, and quantum theory can be formulated by introducing an effective Hamiltonian dependent on clinical content and on a neuromodulatory control vector m_t:

$$\hat{H}(t) = \hat{H}_0 + \sum_r \lambda_r \hat{O}_r(z_t, M_t) + \sum_m u_m(m_t)\hat{M}_m, \qquad \Gamma_t = \text{softplus}(w_\Gamma^\top m_t).$$

Where:

- $\hat{H}_0$ is the system's base term;
- $\hat{O}_r$ are diagnostic observables dependent on latent state z_t and memory M_t;
- λ_r are coupling coefficients;
- m_t is the vector of neuromodulatory surrogates (novelty, volatility, reward prediction, alertness, conflict);
- $\hat{M}_m$ are modulation operators;
- Γ_t is an effective decoherence/commit rate.

Within this framework neuromodulators are not abolished by quantum theory: they become regulators of precision, stability, and transition from superposition to selection.

7.6 Training, new-case generation, and the statistical dream

The requirement to include generative training as an integral part of the model of thought is made explicit here.

7.6.1 Latent statistical cloud

Given a clinical context c composed of symptoms, examinations, physical traits, provenance, therapeutic history, and known patterns, Melampo constructs a conditioned latent cloud:

$$z \sim p_\theta(z \mid c), \qquad \tilde{x} \sim p_\theta(x \mid z, c).$$

Where:

- c is the composite clinical context;
- z is the latent configuration of the statistical dream;
- $\tilde{x}$ is a generated or reorganized synthetic case.

7.6.2 Coherence filter and consolidation

Every synthetic case must be passed through a coherence filter:

$$a(\tilde{x}, c) = \mathbb{1}[S_{\text{coh}}(\tilde{x}, c) \geq \tau_{\text{coh}} \ \wedge \ R(\tilde{x}, c) \leq \tau_R].$$

Where:

- $S_{\text{coh}}(\tilde{x}, c)$ is an anatomo-clinical coherence score;
- $R(\tilde{x}, c)$ is a risk of incoherence or epistemic harm;
- τ_{coh} is the minimum coherence threshold;
- τ_R is the maximum acceptable risk threshold;
- $a(\tilde{x}, c)$ determines whether the synthetic case is accepted.

7.6.3 Clinical meaning of the statistical dream

The statistical dream is not an indiscriminate generator. It is a process of memory reorganization in which the collapse of a diagnostic wavefunction may represent:

- discovery of a new coherent link between distant patterns;
- creation of a useful prototype for rare cases;
- emergence of an incoherent case that is discarded but sharpens the decision boundary;
- a counterfactual useful to the differential diagnosis.

In this sense, generative replay becomes part of the theory of intuition: intuition is not only online decision; it is also the **offline reorganization of the space of hypotheses**.

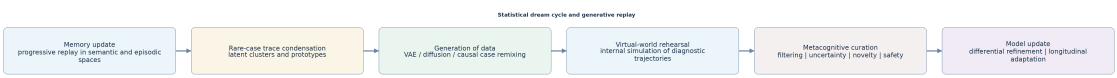

Figure 7. Statistical dream cycle and generative replay.

7.7 Microtubules, Orch-OR, and the biophysical frontier

The microtubular line and Orch-OR-type models remain part of the project as a theoretical-quantum frontier under examination [46–48]. The scientific

attitude adopted here is threefold:

- not to assume as established what has not yet reached empirical consensus;
- not to expel from the project hypotheses with high theoretical content;
- to define criteria for differential testing, replication, and possible falsification.

The same logic applies to the quantum biology literature on radical pairs, the quantum Zeno effect in cryptochrome, NMDA receptor activity, and proton tunneling [42–45]. These results do not prove a strong quantum brain, but they do show that quantum phenomena in biology are not conceptually inadmissible.

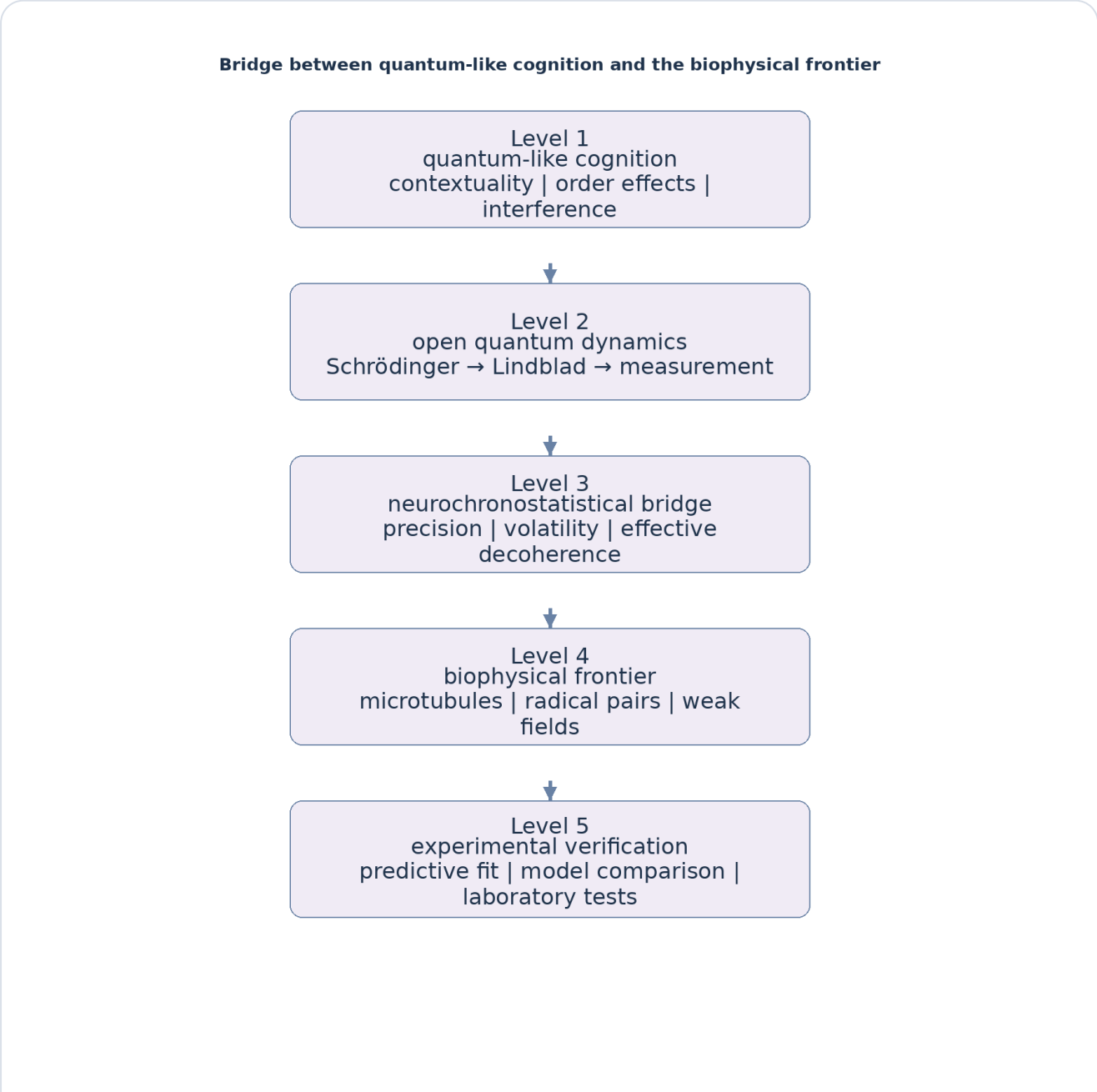

Figure 8. Bridge between quantum-like cognition and the biophysical frontier.

8. Platform for Verification, Validation, and Falsifiability

Melampo's robustness depends on a **multi-level validation platform**. The goal is not to defend the project, but to construct a system in which each block can be confirmed, resized, or rejected.

8.1 Verification of the perceptual and multimodal core

First level: technical benchmarks for segmentation, detection, report drafting, retrieval, and image-text coherence. Minimum endpoints must include site-wise performance, device-wise performance, subgroup performance, and performance stratified by data quality.

8.2 Verification of reasoning and metacognition

Second level: multi-step clinical reasoning benchmarks of the type developed in 2025–2026, with evaluation not only of final correctness but also of inferential trajectory quality [28–29]. Melampo should be measured on:

- quality of the differential diagnosis;
- correctness of the proposed sequence of questions and tests;
- ability to revise the diagnosis in light of new evidence;
- quality of the rationale and fidelity of the explanation;
- accuracy of abstention.

8.3 Calibration, selective prediction, and out-of-distribution control

Third level: calibration and selective-prediction measures. Here enter Brier score, ECE, NLL, AUROC for OOD detection, risk-coverage curves, false reassurance rate, and escalation latency. Without this level, any claim to clinical intuition remains engineeringly immature.

8.4 Falsifiability program for the theoretical-quantum line

The theoretical-quantum line must be tested on three distinct planes.

1. **Cognitive-formal plane.** Quantum-like models should be benchmarked against Bayesian, energy-based, and other classical state-space baselines on contextuality, order effects, diagnostic reversals, and ambiguity handling.
2. **Dynamical plane.** Schrödinger/Lindblad-type formulations should be tested against classical sequential-inference models on stability, calibration, and the handling of partially conflicting evidence.
3. **Biophysical frontier plane.** Microtubule-, radical-pair-, or other non-classical hypotheses should be treated as separate research tracks with explicit empirical signatures, replication requirements, and a clear rule that no clinical deployment claim may rest on them alone.

8.5 Reader studies, workflow, and human factors

The clinically relevant proof remains: **physician alone versus physician + Melampo**. Recommended endpoints are:

- diagnostic accuracy and differential quality;
- sensitivity to rare cases and weak signals;
- reduction in decision time without increasing risk;
- reduction in false reassurance;
- calibrated clinician trust in the system.

8.6 Governance, regulation, and Total Product Life Cycle

WHO guidance on large multimodal models for health requires attention to transparency, oversight, social risk, responsible use, and continuous governance [34]. In 2025 the FDA consolidated recommendations on Predetermined Change Control Plans for AI-enabled devices, while the IMDRF finalized Good Machine Learning Practice principles [35–37]. In Europe, the AI Act is now part of the compliance landscape for high-risk systems in healthcare [38].

Melampo must therefore be born with:

- dataset cards, model cards, change logs, and audit trails;
- change-control plans and rollback criteria;
- post-deployment drift and bias monitoring;
- re-validation procedures after every substantial model modification;
- segregation between research and clinical environments.

Figure 9. Validation and falsifiability pyramid.

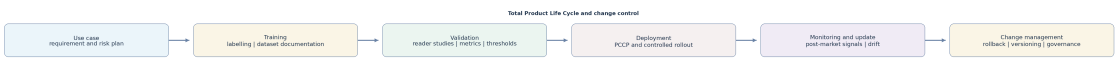

Figure 10. Total Product Life Cycle and change control.

9. Scientific and Technical Roadmap

9.1 Phase I — Verifiable clinical core

- 3D and whole-slide encoders;
- EHR and report parsers;
- KG-RAG memory;
- planner–retriever–generator–verifier orchestrator;
- uncertainty layer with abstention.

9.2 Phase II — Memory and continual learning

- episodic and prototypical memory;
- continual learning with EWC and replay;
- synthetic datasets for rare classes.

9.3 Phase III — Reasoning and metacognition

- adaptive question planning;
- critique loop;
- multi-step reasoning benchmarks;
- self-monitoring and escalation controls.

9.4 Phase IV — Theoretical-quantum track

- contextuality and order-effect benchmarks;
- optional quantum-like layer;
- comparison of Schrödinger/Lindblad formulations with classical baselines;
- design of frontier biophysical protocols.

9.5 Phase V — Clinical validation and responsible industrialization

- reader studies;
- controlled pilot deployment;
- post-market monitoring and change control;
- integration with the regulatory stack and TPLC documentation.

10. Conclusions

Project Melampo can be consolidated without losing its original ambition. The strongest path is to treat it as an integrated scientific architecture in which:

- neurobiology provides functional constraints;
- pedagogy of expertise provides the growth model of intuition;
- multimodal AI provides the operational engine;
- metacognition governs reliability, escalation, and abstention;
- the theoretical-quantum hypothesis remains a living part of the project, but tied to programs of verification and falsification.

The notion of intuition emerging from this treatise is not mystical. It is the **contextual, rapid yet traceable selection of a hypothesis within a space of statistical and semantic possibilities organized by experience, memory, and measurement**. In the fully developed version of the project, the statistical dream, generative replay, quantum-like differential reasoning, and metacognitive control become aspects of the same clinical machine: a machine that learns, corrects itself, abstains when necessary, and accounts for its steps.

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